Literature Review

Smart Irrigation Decision Agent

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1. Introduction

This review looks at three tasks in irrigation. A system must decide how much water to use, check whether its sensor readings are reliable, and share water when the supply is limited. We use the studies to decide how to test those tasks together in Naba.

We chose seven peer-reviewed journal papers published from 2022 to 2026. They cover irrigation rules, sensor faults, and ways to divide water. We used full papers, indexed sections, or publisher previews depending on what we could access.

2. Rules for controlling irrigation

Fuzzy rules let a controller respond to degrees such as slightly dry or very dry. Singh et al. [1] used four inputs and 81 rules to set pump speed. They tested the design in Simulink and on an Arduino. The pump voltage changed as the inputs changed. Their test showed the pump response. It did not compare water use over a growing season.

Neugebauer et al. [2] tested a fuzzy controller against a fixed schedule in a 12-day grass and soil simulation. They tried nine sensor depths. At 235 mm, the controller used 11.5% less water than the fixed schedule. At 910 mm, it used about 20% more. In a separate test, changing the controller settings raised water use by about 38%.

Tabatabaei [3] tested a controller with 27 rules and a rain check in an Iranian vineyard. Over one season, the test plot used 340 m^3/ha and the nearby plot on the usual schedule used 520 m^3/ha. The test plot used 34.6% less water. Each method had one plot, so differences between the plots may have affected the result.

What the comparison tells us

Sensor depth and rule settings can change whether a controller saves water [2]. The field test [3] adds results from a real growing season. Both studies focus on the field or plot. A shared supply adds another question. If one farm gets more water because of a bad reading, how much less can the others receive?

Sensor checks and allocation

3. Checking unreliable readings

Jihani et al. [4] used past readings and a Kalman filter to predict the next reading. They flagged a fault when the reading moved too far from the prediction. They tested sudden faults in a lab with two sensor nodes. Their test measured fault detection. We also need to measure what happens to water use when a reading is rejected.

Our agent predicts soil moisture using the previous estimate, the water applied, and the expected water loss. It uses this prediction when it rejects a reading. If a sensor fails for several days, small errors in the expected water loss can build up.

4. Allocating limited water

Shaikh et al. [5] used two linear programming models to share water among canal branches in Pakistan during low flows. First, they used the monthly supply to choose crop areas. Then they used the seven-day supply to set watering levels. They reported an allocation that avoided conflicts. The preview describes canal planning and does not report daily sensor checks.

Bulukazari et al. [6] used AquaCrop and an optimization model to plan land and water use in Iran. They tested three soil salt levels, three weather conditions, and five irrigation strategies. Their model estimated an average of 20% less water at the same profit or 18% more profit at the same water use. The results depend on the model and its inputs.

Chang et al. [7] modeled the daily movement of soil water and groundwater in five areas of an irrigation district. They used fuzzy and interval optimization to compare fairness, economic returns, and water efficiency when inputs were uncertain. The paper preview does not report tests with missing or stuck sensor readings.

How the studies inform our design

The irrigation rules in [1]-[3] support the knowledge pillar. The prediction check in [4] helps us handle uncertainty when readings are unreliable. The shared water models in [5]-[7] guide our optimization goal and the limits on each water share.

We check that the total water used stays within the daily supply. We also measure the largest daily moisture shortage and count how often farms stay at or above the level where crop stress begins. These measures show whether protecting the driest farm helps the group as a whole.

Comparison of the evidence

5. Critical comparison

The studies measure different things, including pump response, water use, fault detection, and profit. Table 1 shows what each test can tell us.

Table 1. Summary of the reviewed studies.

Ref	AI technique	Problem / test cases	Evaluation and result	Limitation
[1]	Mamdani fuzzy rules	Pump test in Simulink and on Arduino	81 rules set voltage from four inputs. The pump responded to changes in the inputs.	Test covered pump response only.
[2]	Fuzzy rules and soil simulation	Grass simulation with nine sensor depths	11.5% less water at 235 mm. About 20% more at 910 mm.	Test used simulated soil and changed the sensor depth.
[3]	Fuzzy rules and a rain check	Two nearby vineyard plots over one season	340 m^3/ha in the test plot and 520 m^3/ha in the comparison plot.	One plot per method.
[4]	Time-series model and Kalman filter	Sudden faults in a lab with two sensor nodes	Flagged faults by comparing readings with predictions.	Measured fault detection only.
[5]	Linear programming	Canal branches with low flow	Monthly crop plan and seven-day watering plan. Reported no conflicts.	Preview covers canal planning.
[6]	Nonlinear optimization and AquaCrop	Water and land plans for a region	20% less water at the same profit or 18% more profit at the same water use.	Estimates from a planning model.
[7]	Fuzzy and interval optimization	Daily water balance in five areas	Compared fairness, economic returns, and water efficiency.	Preview gives no tests of missing or stuck readings.

Limits of the evidence

Detecting a fault only tells us the reading was flagged. To see whether that helps crops, we need to check how much water each farm received and how wet the soil was at the end of the day. A test with several farms also shows how one farm's water share affects the others.

Research gap and references

6. Research gap

The reviewed papers cover irrigation control [1]-[3], fault detection [4], and shared water planning [5]-[7]. Chang et al. already model daily water use [7]. In the studies we reviewed, we found no test that compares different ways to handle faulty readings when several farms share a daily water limit.

Naba runs the same six farms with three strategies. One checks readings, one uses a fixed schedule, and one uses readings without checking for faults. The two strategies that use sensors share the same rules for dividing water and the same estimate for missing readings. We give all three the same starting soil, weather, and faults. This lets us see what the checks change, including when the water supply drops.

References

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[5] I. A. Shaikh *et al.*, “Optimizing approach of water allocation to off-takes during reduced flows,” *Water Resour. Manage.*, vol. 36, no. 3, pp. 891-913, 2022, doi: 10.1007/s11269-021-03054-4.

[6] S. Bulukazari, H. Babazadeh, N. Ebrahimipak, S.-H. Mousavi-Jahromi, and H. Ramezani Etedali, “Optimization of water and land allocation in salinity and deficit-irrigation conditions at farm level in Qazvin plain,” *PLOS ONE*, vol. 17, no. 7, Art. no. e0269663, 2022, doi: 10.1371/journal.pone.0269663.

[7] H. Chang, G. Li, C. Zhang, and Z. Huo, “Efficient and equitable irrigation management: A fuzzy multi-objective optimization model integrating water movement processes,” *J. Environ. Manage.*, vol. 371, Art. no. 123164, 2024, doi: 10.1016/j.jenvman.2024.123164.

Source access. We checked the full texts of [1]-[3] and [6]. For [4], we checked the indexed methods and experiment sections. For [5] and [7], we checked publisher previews and abstracts.